

Damage ODE verification: uniaxial displacement-controlled test

backward Euler against an independent scalar-ODE reference; error decays after the activation kink rather than compounding

stress-damage coupling checked algebraically at every quadrature point: $\max|\sigma_{zz}^{\text{FE}} - \sigma_{zz}^{\text{alg}}| = 0.00361 \text{ Pa} = 1.8\text{e-}11 \sigma_{th}$ (round-off, at LinearSolverTolerance 10^{-12})

